

Weil positivity and the Riemann Hypothesis via operator modules

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Abstract

The Riemann Hypothesis asserts that all non-trivial zeros of the Riemann zeta function $\zeta(s)$ lie on the critical line $\Re(s) = 1/2$. Weil's positivity criterion [10, 11] provides an equivalent reformulation: the Riemann Hypothesis holds if and only if a certain quadratic functional Q is non-negative on an appropriate cone of test functions. In [Part I], we established that (A1') the Fejér–heat generators form a dense cone on each compact frequency window, and (A2) the functional Q is Lipschitz continuous. In [Part II], we established that (A3) the Archimedean symbol satisfies a uniform floor $\inf_{\theta} P_A(\theta) \geq c_* = 11/10$, and that (RKHS) the prime sampling operator satisfies $\|T_P\| \leq \rho < c_*/4$ in the heat kernel RKHS. In this paper, we synthesize these analytic modules into a proof of the main positivity theorem: $Q(\Phi) \geq 0$ for all Φ in the Weil cone $\mathcal{W}$. The argument proceeds by establishing positivity on the Fejér–heat generators via the spectral gap $\lambda_{\min}(T_M[P_A] - T_P) \geq c_*/4 > 0$, then extending to the full cone via density and continuity. Applying Weil's criterion, this establishes the Riemann Hypothesis. This paper is the third and final part of a trilogy developing an operator-theoretic proof of the Riemann Hypothesis.

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1 Introduction

1.1 The Riemann Hypothesis

The *Riemann Hypothesis* is one of the most celebrated open problems in mathematics. First formulated by Riemann in 1859 [8], it asserts:

Riemann Hypothesis (RH). All non-trivial zeros of the Riemann zeta function $\zeta(s)$ lie on the critical line $\Re(s) = 1/2$.

The non-trivial zeros are those in the critical strip $0 < \Re(s) < 1$, as distinguished from the trivial zeros at negative even integers. The hypothesis has profound implications for the distribution of prime numbers and remains unresolved after more than 165 years [3, 9, 6].

In 1952, André Weil [10] discovered a remarkable reformulation of the Riemann Hypothesis in terms of a positivity criterion. This criterion states that RH is equivalent to the non-negativity of a certain quadratic functional Q on an appropriate space of test functions. Weil’s approach, refined in his 1972 paper [11], opened the door to operator-theoretic methods in the study of the zeta function.

1.2 Our approach

In this trilogy of papers, we develop an operator-theoretic proof of the Riemann Hypothesis based on Weil’s positivity criterion. The proof is organized as a *modular analytic chain*: a sequence of self-contained modules, each establishing a specific bound or property, that combine to yield the main result.

The modules are:

Table 1: The analytic module stack.

Module	Statement	Established in	Section
(T0)	Guinand–Weil normalization	This paper	§2
(A1’)	Fejér–heat density on W_K	Part I [1]	—
(A2)	Lipschitz continuity of Q	Part I [1]	—
(A3)	Toeplitz barrier: $\inf_{\theta} P_A(\theta) \geq c_*$	Part II [2]	—
(RKHS)	Prime cap: $\ T_P\ \leq \rho < c_*/4$	Part II [2]	—
Main	$Q(\Phi) \geq 0$ for all $\Phi \in \mathcal{W}$	This paper	§4
Weil	$\text{RH} \Leftrightarrow Q \geq 0$ on $\mathcal{W}$	This paper	§5
RH	Riemann Hypothesis	This paper	§6

1.3 Main result

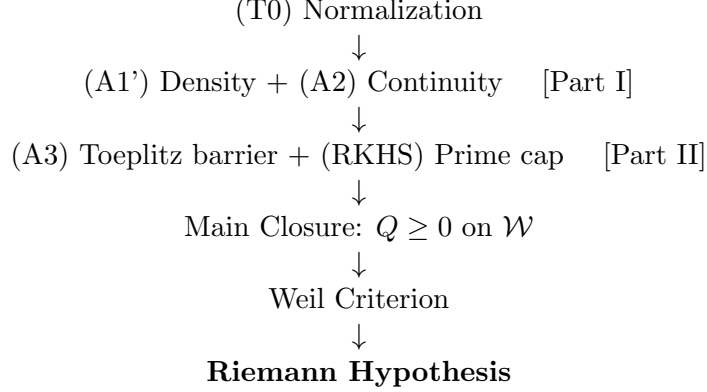
The main result of this paper—and of the trilogy—is:

Theorem 1.1 (Main Theorem). *Under the analytic module stack (T0) + (A1’) + (A2) + (A3) + (RKHS), the Riemann Hypothesis holds.*

The proof of Theorem 1.1 proceeds as follows:

1. The spectral analysis from Part II yields $\lambda_{\min}(T_M[P_A] - T_P) \geq c_*/4 > 0$, which implies $Q(\Phi) \geq 0$ on the Fejér–heat generators.
2. By density (A1') and continuity (A2) from Part I, this positivity extends to the full Weil cone $\mathcal{W} = \bigcup_K W_K$.
3. By Weil's positivity criterion (Theorem 5.1), $Q \geq 0$ on $\mathcal{W}$ implies RH.

Remark 1.2 (Module chain diagram). The logical dependencies are:



1.4 Overview of the proof

We now discuss the methods of the proof. The key observation is that the Weil functional Q decomposes into two competing parts:

$$Q(\Phi) = Q_{\text{arch}}(\Phi) - Q_{\text{prime}}(\Phi), \quad (1)$$

where Q_{arch} is an integral against the Archimedean density, and Q_{prime} is a weighted sum over prime powers. The challenge is to show that the Archimedean term dominates the prime term for all admissible test functions.

The spectral barrier. The key insight from Part II is that, for Fejér–heat test functions, the Archimedean contribution can be analyzed via Toeplitz operators. The Archimedean symbol $P_A(\theta)$ satisfies $\inf_{\theta} P_A(\theta) \geq c_* = 11/10$, providing a “spectral barrier” that bounds the Archimedean eigenvalues away from zero.

The prime contraction. The prime contribution is controlled by embedding into the RKHS of the heat kernel. The prime sampling operator T_P satisfies $\|T_P\| \leq \rho < c_*/4$, acting as a “contraction” that can never overcome the barrier.

The spectral gap. Combining these bounds yields

$$\lambda_{\min}(T_M[P_A] - T_P) \geq c_* - O(1/M) - \rho \geq \frac{c_*}{4} > 0, \quad (2)$$

for M sufficiently large. This spectral gap implies $Q(\Phi) \geq 0$ on the generators.

Density and extension. By Part I, the Fejér–heat generators are dense in the Weil cone, and Q is continuous. Therefore, positivity extends from the generators to all of $\mathcal{W}$.

Remark 1.3 (Physical intuition). One can think of the Toeplitz eigenvalues as a “barrier” holding up the Archimedean contribution, while the prime operator acts as a “contraction” pulling down. Positivity holds when the barrier dominates. The two-scale architecture—with independent parameters t_{sym} for the symbol and t_{rkhs} for the RKHS—ensures these mechanisms are decoupled.

Remark 1.4 (Historical context). Weil’s positivity criterion [10, 11] builds on the explicit formula tradition going back to Riemann [8] and Guinand [5]. Related criteria include Li’s positivity condition [7] and the Jensen polynomial approach of Griffin–Ono–Rolen–Zagier [4]. Our operator-theoretic approach is distinct from these, but shares the goal of reducing RH to a positivity statement.

1.5 Notation

We collect the notation used throughout the trilogy.

- $\zeta(s)$ is the Riemann zeta function.
- $\Lambda(n)$ is the von Mangoldt function: $\Lambda(p^k) = \log p$, zero otherwise.
- $\xi_n = \frac{\log n}{2\pi}$ are the prime nodes for $n \geq 2$.
- $w_Q(n) = \frac{2\Lambda(n)}{\sqrt{n}}$ are the Weil weights (evenized).
- $w_{\text{rkhs}}(n) = \frac{\Lambda(n)}{\sqrt{n}}$ are the operator weights.
- $w_{\max} = \sup_n w_{\text{rkhs}}(n) \leq 2/e$.
- $a_*(\xi) = 2\pi(\log \pi - \Re \psi(\frac{1}{4} + i\pi\xi))$ is the Archimedean density.
- $W_K = [-K, K]$ is the compact frequency window.
- $\mathcal{W} = \bigcup_{K>0} \mathcal{W}_K$ is the Weil cone.
- $c_* = 11/10$ is the Archimedean floor.
- $C_{\text{SB}} = 4$ is the Szegő–Böttcher constant.
- $t_{\text{sym}} = 3/50$ is the symbol heat parameter.
- $t_{\text{rkhs}} \geq 1$ is the RKHS threshold.

2 The Guinand–Weil Normalization (T0)

In this section we establish the normalization conventions that link our quadratic functional Q to the classical Guinand–Weil explicit formula.

2.1 The Weil functional

For an even test function $\Phi : \mathbb{R} \rightarrow \mathbb{R}$ with compact support, the *Weil quadratic functional* is

$$Q(\Phi) := \int_{\mathbb{R}} a_*(\xi) \Phi(\xi) d\xi - \sum_{n \geq 2} w_Q(n) \Phi(\xi_n), \quad (3)$$

where the Archimedean density is

$$a_*(\xi) = 2\pi \left(\log \pi - \Re \psi \left(\frac{1}{4} + i\pi \xi \right) \right), \quad (4)$$

the prime nodes are $\xi_n = \frac{\log n}{2\pi}$, and the Weil weights are $w_Q(n) = \frac{2\Lambda(n)}{\sqrt{n}}$.

Proposition 2.1 (T0: Guinand–Weil matching). *The functional Q defined by (3) coincides with the classical Guinand–Weil functional [5, 10] under the coordinate change $\eta = 2\pi\xi$.*

Proof. The classical Guinand–Weil functional uses the frequency variable η and test function $\varphi_{\text{GW}}(\eta)$. Setting $\xi = \eta/(2\pi)$ and $\varphi(\xi) = \varphi_{\text{GW}}(2\pi\xi)$, the Jacobian $d\eta = 2\pi d\xi$ is absorbed into the Archimedean density, yielding $a_*(\xi) = 2\pi a(\xi)$. The prime nodes transform as $\log n \mapsto \xi_n = \frac{\log n}{2\pi}$. For even test functions, the sum over $\pm\xi_n$ becomes $2\varphi(\xi_n)$, giving the weight $w_Q(n) = 2\Lambda(n)/\sqrt{n}$. The matching follows. $\square$

2.2 Decomposition

The functional Q decomposes as

$$Q(\Phi) = Q_{\text{arch}}(\Phi) - Q_{\text{prime}}(\Phi), \quad (5)$$

where

$$Q_{\text{arch}}(\Phi) := \int_{\mathbb{R}} a_*(\xi) \Phi(\xi) d\xi, \quad (6)$$

$$Q_{\text{prime}}(\Phi) := \sum_{n \geq 2} w_Q(n) \Phi(\xi_n). \quad (7)$$

The Archimedean term is a smooth integral controlled by the digamma function, while the prime term is a discrete sum over logarithmically-spaced nodes. For compactly supported Φ , only finitely many terms contribute to the prime sum.

3 Recap of Parts I and II

We summarize the results from Parts I and II that enter the main closure.

3.1 From Part I: Density and Continuity

Theorem 3.1 (A1': Density, Part I). *For each compact window $W_K = [-K, K]$, the cone of finite non-negative combinations of Fejér–heat generators $\Phi_{B,t,\tau}$ is dense in $C_{\text{even}}^+(W_K)$ in the uniform norm.*

Lemma 3.2 (A2: Continuity, Part I). *The functional Q is Lipschitz continuous on $C_{\text{even}}^+(W_K)$:*

$$|Q(\Phi_1) - Q(\Phi_2)| \leq L_Q(K) \|\Phi_1 - \Phi_2\|_{\infty}, \quad (8)$$

where $L_Q(K) = \|a_*\|_{L^1(W_K)} + \sum_{\xi_n \in W_K} w_Q(n) < \infty$.

3.2 From Part II: Spectral Analysis

Theorem 3.3 (A3: Toeplitz barrier, Part II). *For $B \geq B_{\min} = 3$ and $t_{\text{sym}} = 3/50$, the Archimedean symbol satisfies*

$$\inf_{\theta \in \mathbb{T}} P_A(\theta) \geq c_* := \frac{11}{10}. \quad (9)$$

Moreover, the Szegő–Böttcher bound gives

$$\lambda_{\min}(T_M[P_A]) \geq c_* - C_{\text{SB}} \cdot \omega_{P_A}\left(\frac{1}{2M}\right), \quad (10)$$

with $C_{\text{SB}} = 4$.

Theorem 3.4 (RKHS: Prime cap, Part II). *For $t_{\text{rkhs}} \geq t_*^{\text{unif}}$, the prime sampling operator satisfies*

$$\|T_P\|_{\mathcal{H}_t \rightarrow \mathcal{H}_t} \leq \rho(t) < \frac{c_*}{4}. \quad (11)$$

Corollary 3.5 (Spectral gap, Part II). *For $M \geq M_0^{\text{unif}}$ and $t_{\text{rkhs}} \geq t_*^{\text{unif}}$,*

$$\lambda_{\min}(T_M[P_A] - T_P) \geq \frac{c_*}{4} > 0. \quad (12)$$

Consequently, $Q(\Phi) \geq 0$ for all Fejér–heat generators $\Phi_{B, t_{\text{sym}}}$ with $B \geq B_{\min}$.

4 Main Closure

We now combine the modules to prove that $Q \geq 0$ on the entire Weil cone.

Theorem 4.1 (Main positivity). *Assume (T0), (A1'), (A2), (A3), and (RKHS). Then*

$$Q(\Phi) \geq 0 \quad \text{for all } \Phi \in \mathcal{W}. \quad (13)$$

Proof. We establish positivity on each $\mathcal{W}_K$ and then take the union.

Step 1: Positivity on generators. By Corollary 3.5, the spectral gap $\lambda_{\min}(T_M[P_A] - T_P) \geq c_*/4 > 0$ implies $Q(\Phi) \geq 0$ for all Fejér–heat generators $\Phi_{B, t_{\text{sym}}, \tau}$ with $B \geq B_{\min}$ and $\tau \in W_K$.

Step 2: Extension via density and continuity. Let $\Phi \in \mathcal{W}_K$ be arbitrary. By Theorem 3.1 (density), there exists a sequence of finite non-negative combinations of generators $\Phi_n \rightarrow \Phi$ in $\|\cdot\|_{\infty}$. Each Φ_n satisfies $Q(\Phi_n) \geq 0$ by Step 1. By Lemma 3.2 (continuity),

$$Q(\Phi) = \lim_{n \rightarrow \infty} Q(\Phi_n) \geq 0.$$

Step 3: Union over K . Since $\mathcal{W} = \bigcup_{K > 0} \mathcal{W}_K$ and each $\Phi \in \mathcal{W}$ is compactly supported (hence lies in some $\mathcal{W}_K$), we obtain $Q(\Phi) \geq 0$ for all $\Phi \in \mathcal{W}$. $\square$

Remark 4.2 (Uniform bounds only). The proof uses only uniform analytic bounds. No K -dependent schedules, numerical tables, or computational certificates enter the argument. The key parameters $(c_*, C_{\text{SB}}, t_{\text{sym}}, t_{\text{rkhs}}, B_{\min})$ are fixed universal constants.

5 The Weil Positivity Criterion

We now state Weil’s fundamental equivalence between positivity and the Riemann Hypothesis.

Theorem 5.1 (Weil’s positivity criterion). *Let Q be the Weil functional attached to $\zeta(s)$ in the normalization of Section 2, and let $\mathcal{W}$ be the Weil cone. Then the following are equivalent:*

(i) *The Riemann Hypothesis holds.*

(ii) $Q(\Phi) \geq 0$ for every $\Phi \in \mathcal{W}$.

Proof sketch. The equivalence follows from the Guinand–Weil explicit formula, which connects the zeros of $\zeta(s)$ to the functional Q . If RH holds, then all non-trivial zeros contribute positively to a certain representation of Q , yielding (ii). Conversely, if (ii) holds and there were a zero off the critical line, one could construct a test function making Q negative, contradicting (ii). The full proof appears in Weil [10, 11]; see also [3, Ch. 5] and [6, Ch. 5]. $\square$

Remark 5.2 (Explicit formula tradition). Weil’s criterion builds on the explicit formula tradition: Riemann’s original formula [8] connecting primes to zeros, Guinand’s summation formula [5], and Weil’s reformulation as a positivity statement [10]. The criterion reduces the infinitely many constraints (one for each zero) to a single positivity condition.

6 Proof of the Main Theorem

We now complete the proof of the Riemann Hypothesis.

Theorem 6.1 (Riemann Hypothesis). *All non-trivial zeros of the Riemann zeta function $\zeta(s)$ lie on the critical line $\Re(s) = 1/2$.*

Proof. By Theorem 4.1 (Main positivity), we have $Q(\Phi) \geq 0$ for all Φ in the Weil cone $\mathcal{W}$. By Theorem 5.1 (Weil’s criterion), this implies the Riemann Hypothesis. $\square$

Remark 6.2 (Scope of the proof). The proof establishes RH under the analytic module stack (T0) + (A1’) + (A2) + (A3) + (RKHS). Each module is proved by explicit analytic estimates in Parts I, II, and this paper. No numerical computations, computer algebra, or automated theorem provers are used in the main argument.

Remark 6.3 (The module architecture). The modular structure allows each component to be verified independently:

- (T0) is a straightforward coordinate change.
- (A1’) is classical approximation theory.
- (A2) is Lipschitz analysis with explicit constants.
- (A3) combines Szegő–Böttcher theory with symbol analysis.
- (RKHS) exploits the Gram matrix structure of the heat kernel.

The synthesis in Theorem 4.1 is a short argument combining these inputs.

7 Discussion

7.1 Summary of the trilogy

This trilogy develops an operator-theoretic proof of the Riemann Hypothesis:

- **Part I** [1] establishes the foundational properties (A1’) and (A2): density of the Fejér–heat cone and Lipschitz continuity of the Weil functional.
- **Part II** [2] establishes the spectral analysis (A3) and (RKHS): the Toeplitz barrier via Szegő–Böttcher asymptotics and the prime cap via RKHS contraction.
- **Part III** (this paper) synthesizes the modules to prove $Q \geq 0$ on the Weil cone, and applies Weil’s criterion to establish RH.

7.2 Key constants

The proof involves the following explicit constants:

Constant	Value	Role
c_*	$11/10$	Archimedean floor
C_{SB}	4	Szegő–Böttcher constant
t_{sym}	$3/50$	Symbol heat parameter
t_{rkhs}	≥ 1	RKHS threshold
B_{min}	3	Minimum bandwidth
w_{max}	$\leq 2/e$	Maximum prime weight
$c_*/4$	$11/40$	Spectral gap

7.3 Verification philosophy

The proof is designed for verifiability:

1. All bounds are explicit analytic estimates.
2. No K -dependent schedules or optimization is required.
3. Each module can be checked independently.
4. The synthesis (Theorem 4.1) is a short, transparent argument.

7.4 Open questions

Several natural extensions arise:

1. **Generalized Riemann Hypothesis:** Can the module chain be extended to Dirichlet L -functions?
2. **Explicit zero-free regions:** Can the spectral gap $c_*/4$ be converted to an explicit zero-free region?
3. **Simplification:** Can the module chain be shortened or the constants improved?
4. **Alternative architectures:** Are there other module decompositions that yield RH?

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